

ASSIGNMENT NO.3: Conditional Queries and Data Filtering

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Batch	SY. BSc Data Science	Topic	WB(Assignment-3)

Q1) Create a database named CollegeDB and a table named Student.

The table should contain the following columns:

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Age	INT	
Course	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	
Email	VARCHAR(60)	

Insert at least 10 records into the table. Some records should contain NULL values in the Email column.

QUERY:

```
CREATE DATABASE JuniorCollegeDB;
```

```
USE JuniorCollegeDB;
```

```
CREATE TABLE Student (
```

```
    StudentID INT PRIMARY KEY,
```

```
    StudentName VARCHAR(50),
```

```
    Age INT,
```

```
    Course VARCHAR(30),
```

```
    City VARCHAR(30),
```

```
    Marks INT,
```

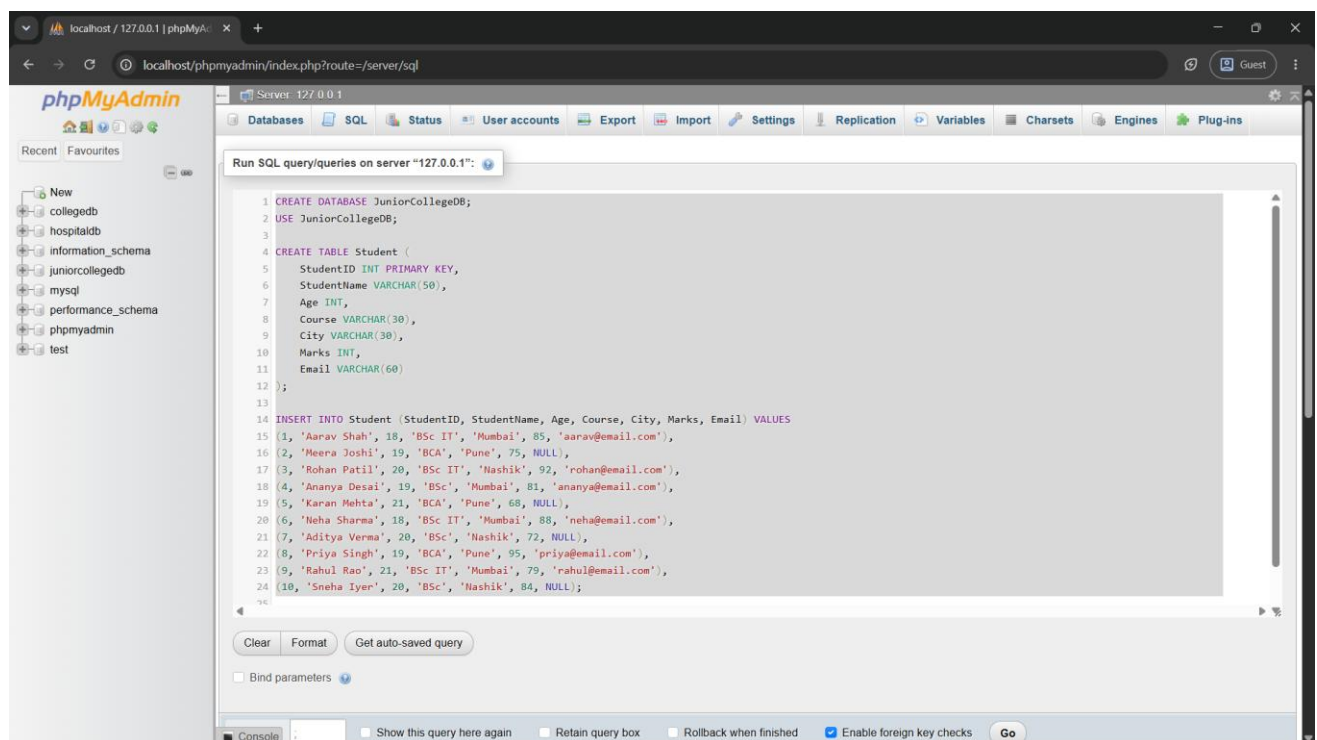
```
    Email VARCHAR(60)
```

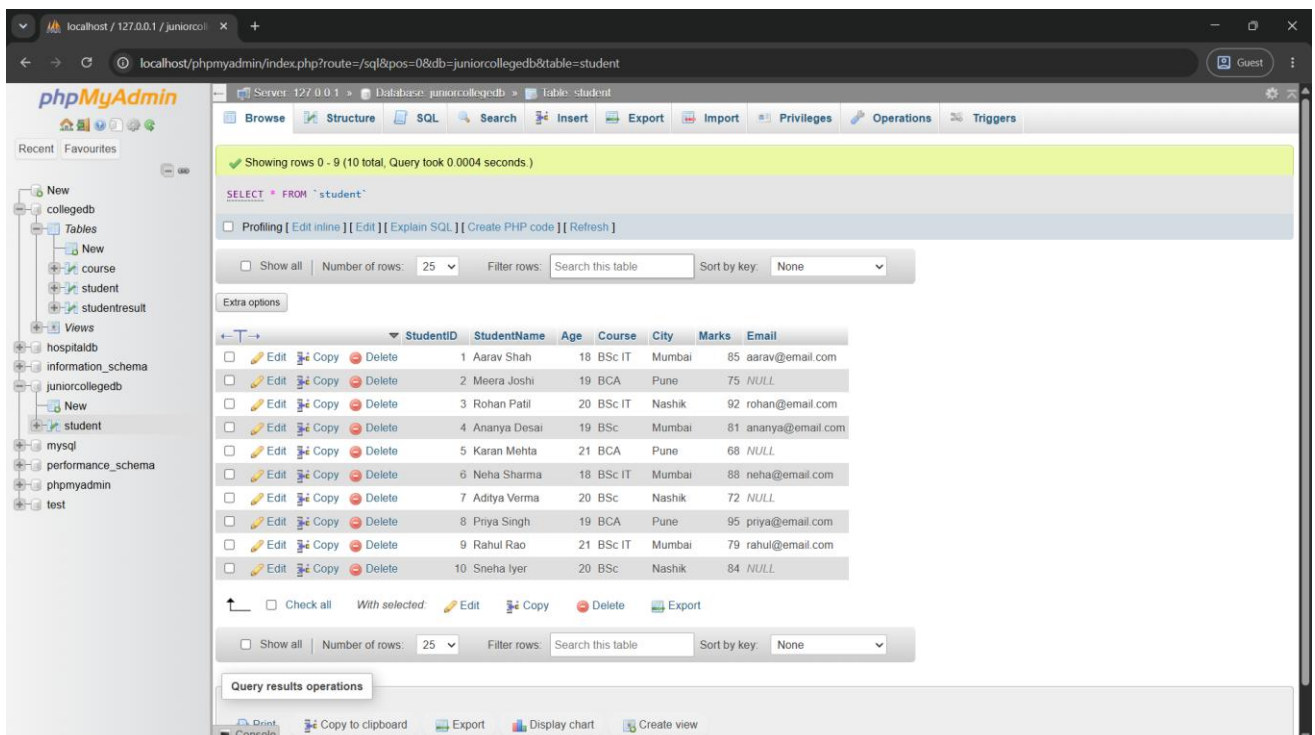
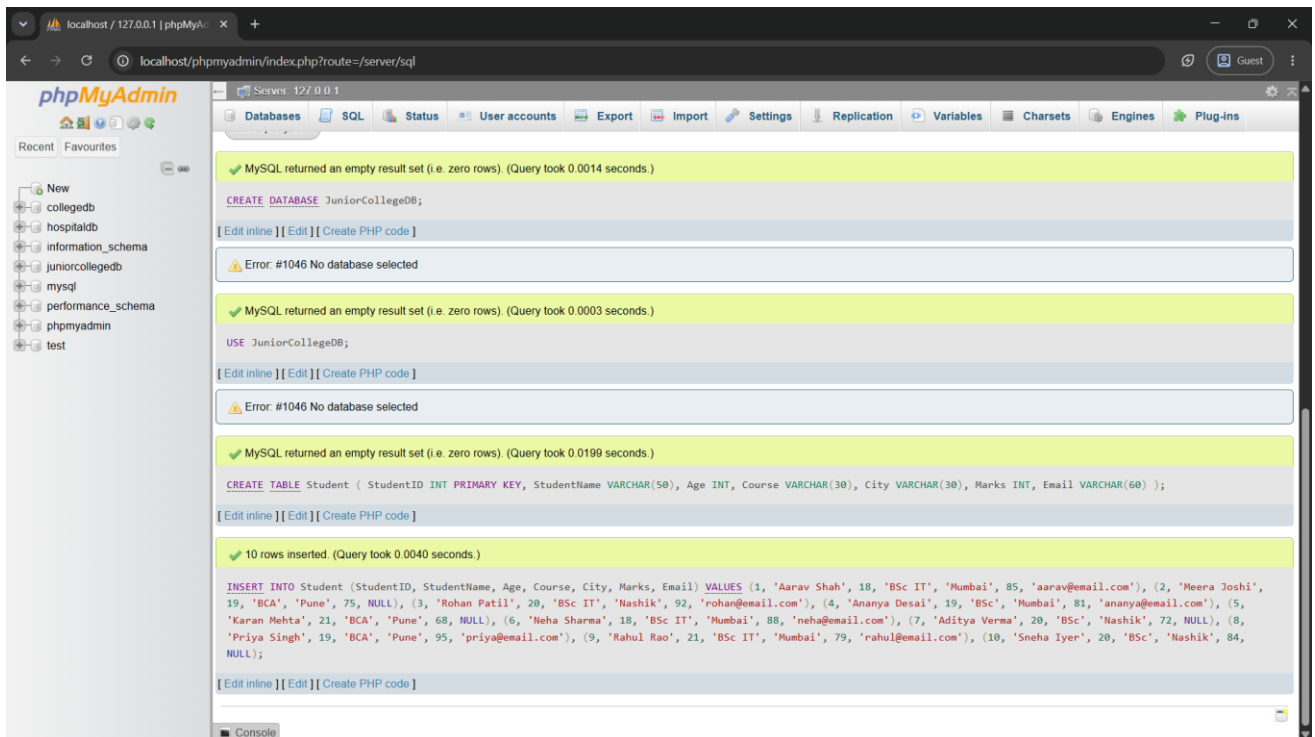
```
);
```

```
INSERT INTO Student (StudentID, StudentName, Age, Course, City, Marks, Email) VALUES
```

(1, 'Aarav Shah', 18, 'BSc IT', 'Mumbai', 85, 'aarav@email.com'),
(2, 'Meera Joshi', 19, 'BCA', 'Pune', 75, NULL),
(3, 'Rohan Patil', 20, 'BSc IT', 'Nashik', 92, 'rohan@email.com'),
(4, 'Ananya Desai', 19, 'BSc', 'Mumbai', 81, 'ananya@email.com'),
(5, 'Karan Mehta', 21, 'BCA', 'Pune', 68, NULL),
(6, 'Neha Sharma', 18, 'BSc IT', 'Mumbai', 88, 'neha@email.com'),
(7, 'Aditya Verma', 20, 'BSc', 'Nashik', 72, NULL),
(8, 'Priya Singh', 19, 'BCA', 'Pune', 95, 'priya@email.com'),
(9, 'Rahul Rao', 21, 'BSc IT', 'Mumbai', 79, 'rahul@email.com'),
(10, 'Sneha Iyer', 20, 'BSc', 'Nashik', 84, NULL);

OUTPUT:





Q2) Write a MySQL query to display students who are from Mumbai and have marks greater than 80.

The output should include:

Column

StudentID
StudentName
City
Marks

QUERY:

SELECT StudentID, StudentName, City, Marks

FROM student

WHERE City = 'Mumbai'

AND Marks > 80;

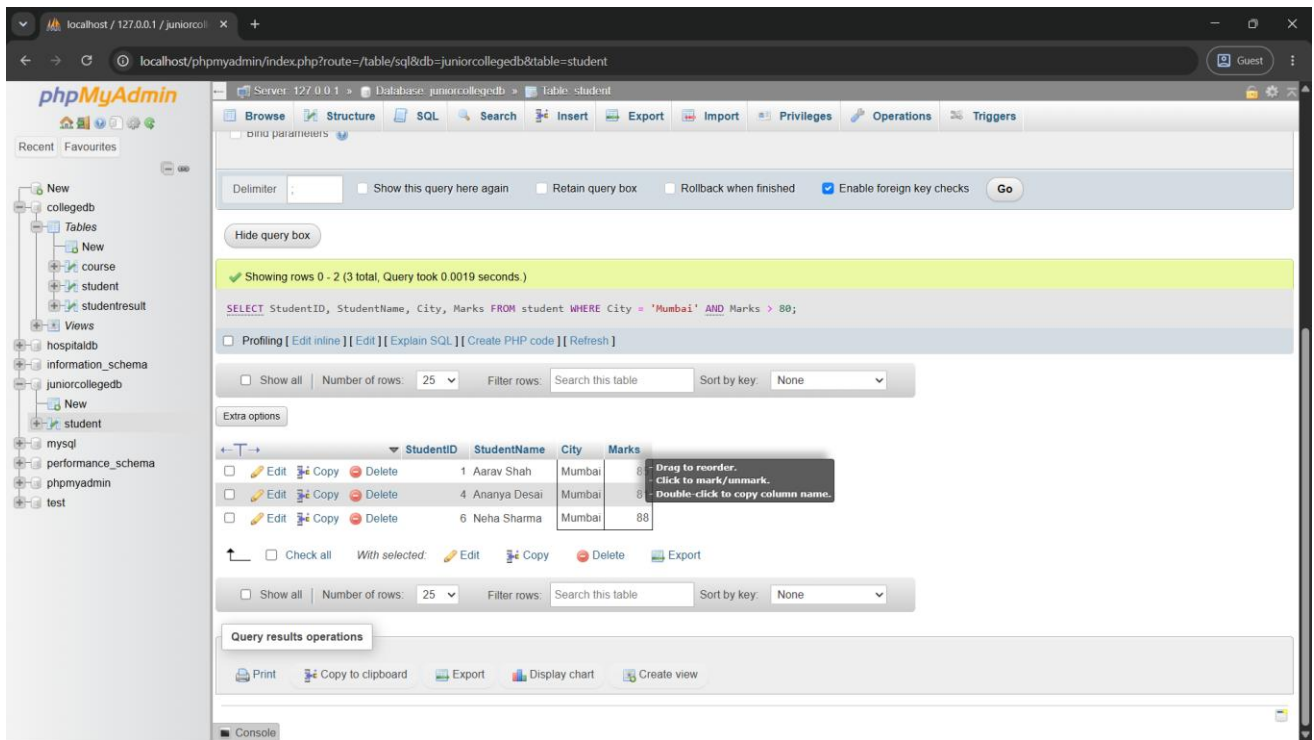
OUTPUT:

The screenshot shows the phpMyAdmin interface for a database named 'juniorcollegedb'. The 'student' table is selected. The SQL query entered is:

```
1 SELECT StudentID, StudentName, City, Marks
2 FROM student
3 WHERE City = 'Mumbai' AND Marks > 80;
```

The query has been executed successfully, showing 2 rows. The results are displayed in a table with the following columns: StudentID, StudentName, City, and Marks. The results are:

StudentID	StudentName	City	Marks
1	John Doe	Mumbai	85
2	Jane Smith	Mumbai	90



Q3) Write a MySQL query to display students who belong to either Pune or Nashik.

Use the OR operator and display the following columns:

Column
StudentName
Course
City

QUERY:

```
SELECT StudentName, Course, City
FROM Student
WHERE City = 'Pune' OR City = 'Nashik';
```

OUTPUT:

localhost / 127.0.0.1 / juniorcol... x +

localhost/phpmyadmin/index.php?route=/table/sql&db=juniorcollegedb&table=student

phpMyAdmin

Recent Favourites

Server: 127.0.0.1 > Database: juniorcollegedb > Table: Student

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table juniorcollegedb.student:

```
1 SELECT StudentName,Course,City
2 FROM Student
3 WHERE City = 'Pune' OR City = 'Nashik';
4
```

StudentID
StudentName
Age
Course
City
Marks
Email

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Hide query box

Showing rows 0 - 5 (6 total, Query took 0.0004 seconds)

```
SELECT StudentName,Course,City FROM Student WHERE City = 'Pune' OR City = 'Nashik';
```

Console [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

localhost / 127.0.0.1 / juniorcol... x +

localhost/phpmyadmin/index.php?route=/table/sql&db=juniorcollegedb&table=student

phpMyAdmin

Recent Favourites

Server: 127.0.0.1 > Database: juniorcollegedb > Table: Student

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Go

Hide query box

Showing rows 0 - 5 (6 total, Query took 0.0004 seconds)

```
SELECT StudentName,Course,City FROM Student WHERE City = 'Pune' OR City = 'Nashik';
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	StudentName	Course	City
☐ Edit ☐ Copy ☐ Delete	Meera Joshi	BCA	Pune
☐ Edit ☐ Copy ☐ Delete	Rohan Patil	BSc IT	Nashik
☐ Edit ☐ Copy ☐ Delete	Karan Mehta	BCA	Pune
☐ Edit ☐ Copy ☐ Delete	Aditya Verma	BSc	Nashik
☐ Edit ☐ Copy ☐ Delete	Priya Singh	BCA	Pune
☐ Edit ☐ Copy ☐ Delete	Sneha Iyer	BSc	Nashik

☐ Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

☐ Print ☐ Copy to clipboard ☐ Export ☐ Display chart ☐ Create view

Console

Q4) Write a MySQL query to display students whose marks are between 70 and 90.

Use the BETWEEN operator and display:

Column
StudentName
Marks

QUERY:

SELECT StudentName,Marks

FROM student

WHERE Marks BETWEEN 70 AND 90;

OUTPUT:

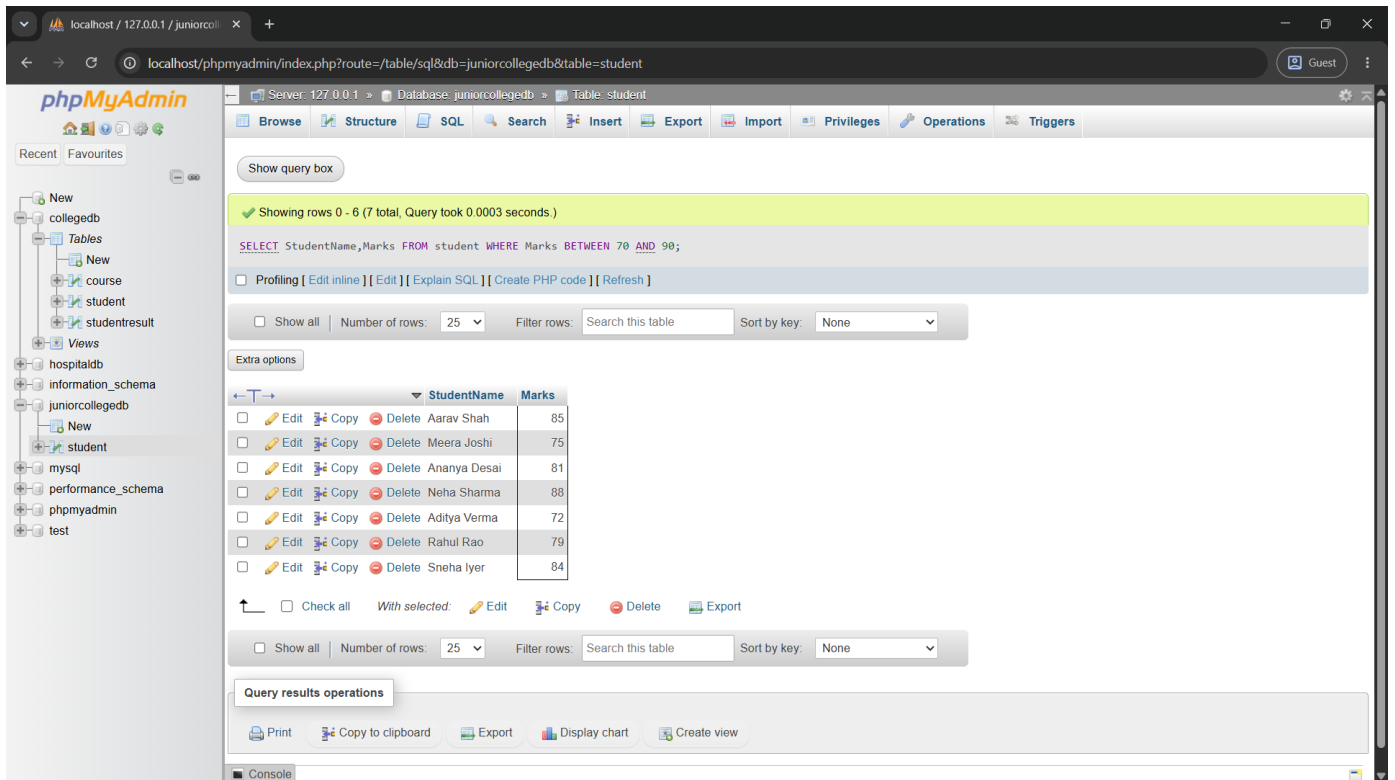
The screenshot shows the phpMyAdmin web interface. The left sidebar displays the database structure, including 'juniorcollegedb' and its tables: 'course', 'student', and 'studentresult'. The main panel shows the 'Table: student' view. A text area contains the SQL query:

```
1 SELECT StudentName,Marks
2 FROM student
3 WHERE Marks BETWEEN 70 AND 90;
4
```

 Below the query area are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. A 'Bind parameters' checkbox is also present. At the bottom, there are options for 'Delimiter', 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks'. A green status bar indicates 'Showing rows 0 - 6 (7 total, Query took 0.0003 seconds)'. Below this, the executed query is shown:

```
SELECT StudentName,Marks FROM student WHERE Marks BETWEEN 70 AND 90;
```

 At the very bottom, there are links for 'Console', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'.



Q5) Write a MySQL query to display students whose names start with the letter A.

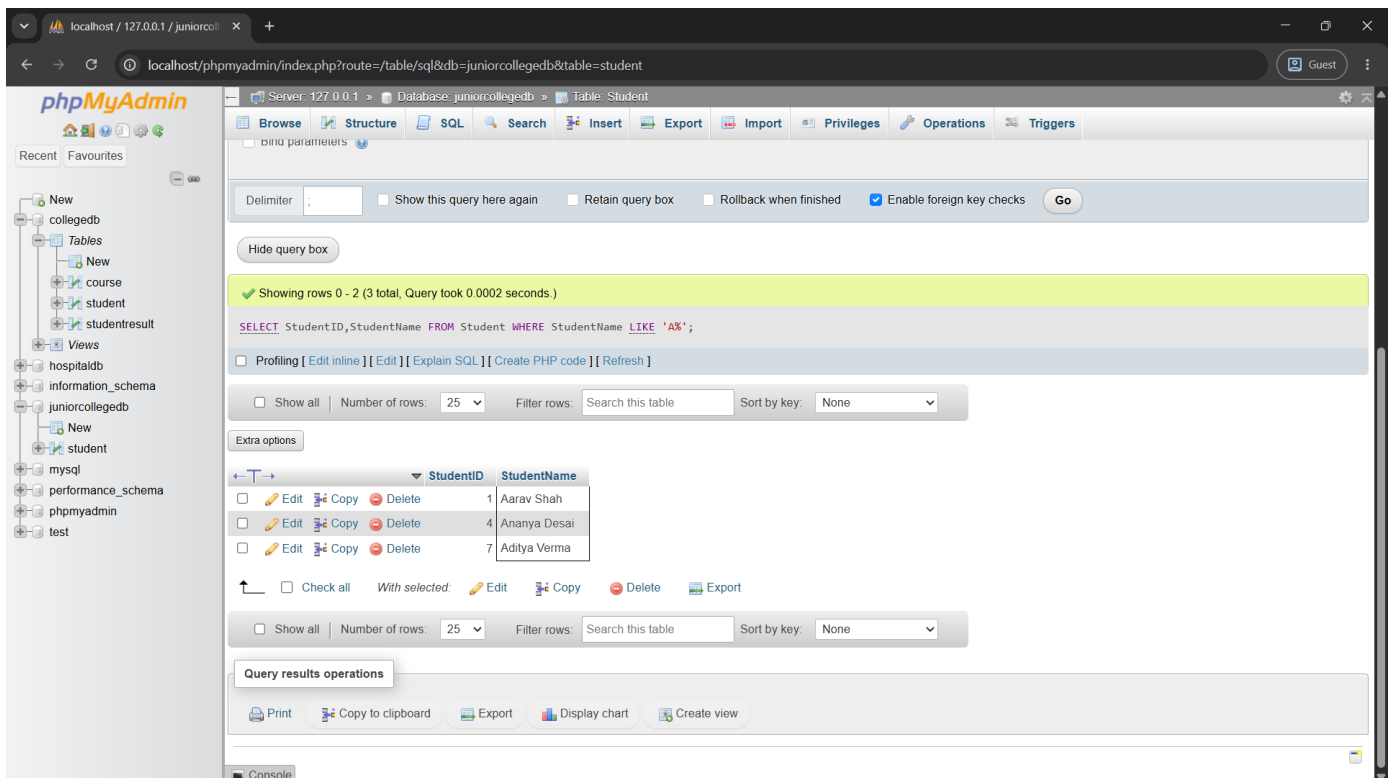
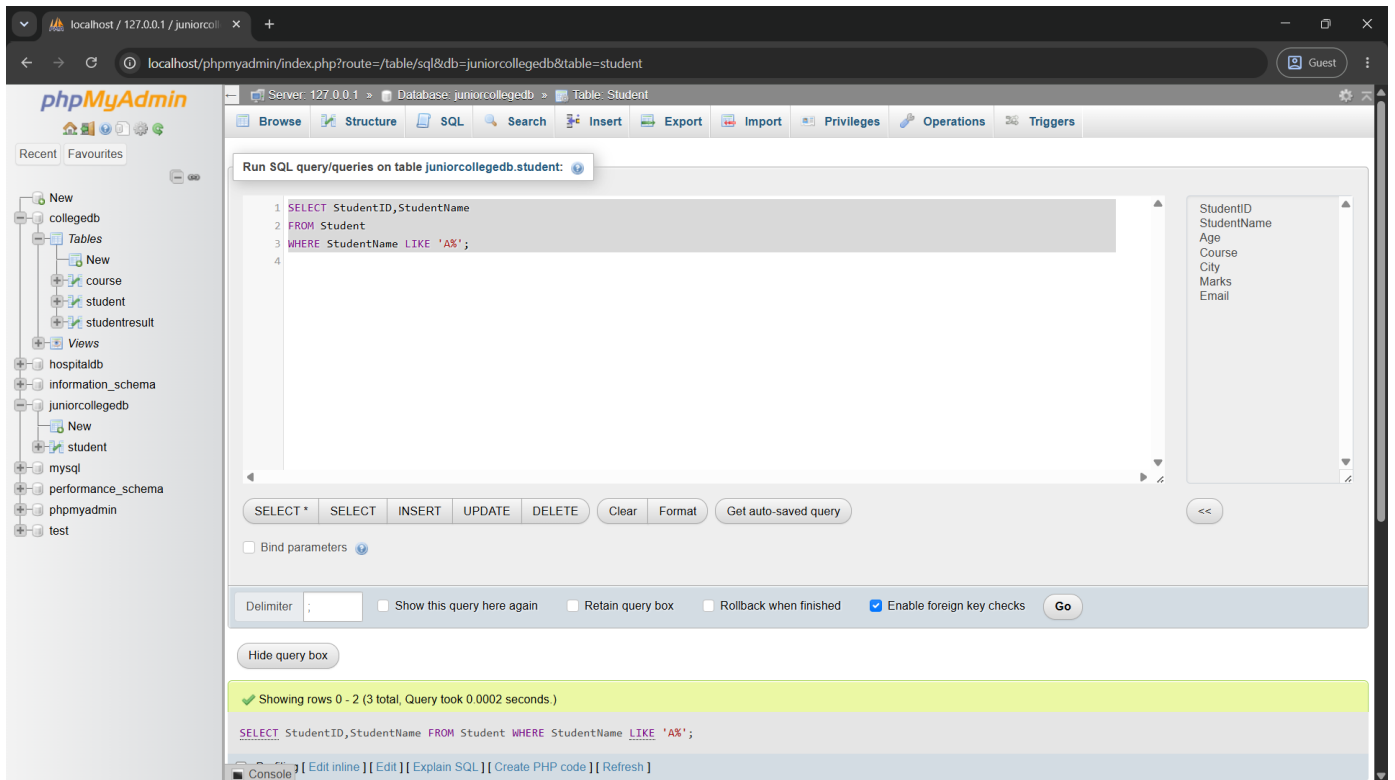
Use the LIKE operator and display:

Column
StudentID
StudentName

QUERY:

```
SELECT StudentID, StudentName
FROM Student
WHERE StudentName LIKE 'A%';
```

OUTPUT:



Q6) Write a MySQL query to display students whose email address is missing.

Use the IS NULL operator and display:

Column
StudentID
StudentName
Email

QUERY:

SELECT StudentID,StudentName,Email

FROM student

WHERE Email IS NULL;

OUTPUT:

The screenshot shows the phpMyAdmin web interface in a browser. The left sidebar displays a database structure with 'juniorcollegedb' selected, showing tables like 'course', 'student', and 'studentresult'. The main panel is titled 'Run SQL query/queries on table juniorcollegedb.student:'. The SQL query entered is:

```
1 SELECT StudentID,StudentName,Email
2 FROM student
3 WHERE Email IS NULL;
4
```

Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and 'Get auto-saved query'. The 'Bind parameters' checkbox is unchecked. At the bottom of the query editor, there are options for 'Delimiter' (set to semicolon), 'Show this query here again', 'Retain query box', 'Rollback when finished', 'Enable foreign key checks' (checked), and a 'Go' button. A 'Hide query box' button is also present.

The execution result is displayed in a green banner: 'Showing rows 0 - 3 (4 total, Query took 0.0002 seconds)'. Below this, the SQL query is repeated, and a 'Console' tab is visible at the bottom.

StudentID	StudentName	Age	Course	City	Marks	Email
1	John Doe	20	Computer Science	New York	85	
2	Jane Smith	21	Mathematics	Los Angeles	78	
3	Mike Johnson	19	Physics	Chicago	92	

localhost / 127.0.0.1 / juniorcolli

localhost/phpmyadmin/index.php?route=/table/sql&db=juniorcollegedb&table=student

Guest

phpMyAdmin

Recent Favourites

New

collegedb

Tables

New

course

student

studentresult

Views

hospitaldb

information_schema

juniorcollegedb

New

student

mysql

performance_schema

phpmyadmin

test

Server: 127.0.0.1 » Database: juniorcollegedb » Table: student

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

Showing rows 0 - 3 (4 total. Query took 0.0002 seconds)

SELECT StudentID,StudentName,Email FROM student WHERE Email IS NULL;

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

StudentID StudentName Email

☐ Edit Copy Delete 2 Meera Joshi NULL

☐ Edit Copy Delete 5 Karan Mehta NULL

☐ Edit Copy Delete 7 Aditya Verma NULL

☐ Edit Copy Delete 10 Sneha Iyer NULL

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Console